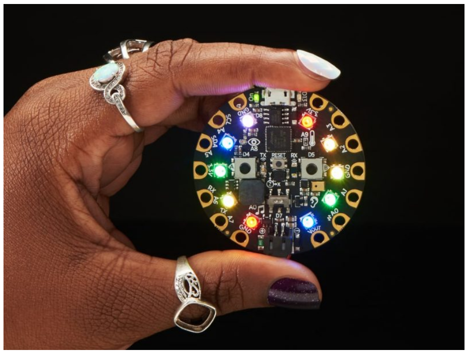
1 kit · 10 boards

Adafruit Circuit Playground Kit

WHAT'S IN THE KIT

- 10 Adafruit Circuit Playground boards
- 10 protective covers
- 10 battery packs



QUICK-START GUIDE

1. Plug a board into a Chromebook/laptop via USB.
2. Go to makecode.adafruit.com and start a new project.
3. Drag blocks to control the 10 built-in LEDs, buttons, and sensors.
4. Click Download and drag the file onto the CPLAYBOOT drive that appears.
5. Unplug, attach the battery pack, and the program runs anywhere!

FULL LESSONS IN THIS GUIDE

PRO TIP: *No installs needed — MakeCode runs fully in the browser, perfect for Chromebooks.*

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LESSON · GRADES 3-5

Mood Ring Machine

LEARNING OUTCOMES

- Students will identify the input sensors and output devices on a microcontroller board
- Students will build a program that reads a sensor and responds with light or sound
- Students will collect and compare simple sensor data across locations

MATERIALS

- Circuit Playground boards with battery packs (1 per pair)
- Chromebooks/laptops with USB ports
- makecode.adafruit.com (free, browser-based)
- Chart paper or Sheets for recording temperature readings

INSTRUCTIONS

1. Before class, plug in one board and test the MakeCode download flow yourself.
2. Warm-up (5 min): pass boards around — ask students to find the buttons, LEDs, and the small sensors.
3. Demo (10 min): build a block program together that turns the LED ring blue when it's cold and red when it's warm.
4. Work time (25 min): pairs build the Mood Ring Machine — LEDs respond to the temperature sensor with at least 3 color ranges.
5. Data walk (10 min): teams carry boards to different spots (window, vent, hallway), record readings, and chart them.
6. Wrap-up: discuss why readings differed and how sensors 'sense.'

ACTIVITY

Mood Ring Machine — students program the LED ring to change color with temperature, then 'read' classmates' moods by fingertip touch. Teams chart readings from around the room and argue from their data about which spots are warmest and why.

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LESSON · GRADES 6-8

Step Counter Challenge

LEARNING OUTCOMES

- Students will use an accelerometer to detect motion events in code
- Students will calibrate a device against a known standard and quantify error
- Students will iterate on threshold values to improve measurement accuracy

MATERIALS

- Circuit Playground boards with battery packs (1 per pair)
- Chromebooks with makecode.adafruit.com
- A real pedometer or phone step-counter for calibration
- Tape or wristbands to wear the boards

INSTRUCTIONS

1. Demo shake-detection blocks and explain thresholds in plain terms: 'how hard is a step?'
2. Pairs build a step counter: on shake → add 1 to a variable → show count on LEDs.
3. Calibration trial 1: walk 50 steps beside a real pedometer; record both counts.
4. Teams adjust their threshold/logic and repeat the 50-step trial at least twice.
5. Collect class data on error rates; discuss which strategies improved accuracy.
6. Exit ticket: one sentence on why identical hardware produced different results.

ACTIVITY

> Step Counter Challenge — wearable step counters built from scratch, tested against a commercial pedometer over 50-step trials. The winning team isn't the first to finish — it's the one whose count comes closest after calibration.